

15.2

(1) $A = a$

$$v = \frac{\omega}{2\pi} = \frac{b}{2\pi}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{b}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{c}$$

$$u = \lambda v = \frac{b}{c}$$

$$\Phi|_{z=0} = a \cos(bt + d)$$

$$\text{当 } t=0 \text{ 时, } \Phi = d$$

(2) $\varphi(z, t) = bt - cz + d$

$$\varphi_1 = bt - cz_1 + d$$

$$\varphi_2 = bt - cz_2 + d$$

$$\Delta\varphi = |\varphi_2 - \varphi_1| = |-c \cdot l| = c \cdot l$$

15.6.

$$\text{由图 } A = 0.4 \text{ m, } u = 0.08 \text{ m/s,}$$

$$\lambda = 2 \times 0.1 = 0.2 \text{ m.}$$

$$\text{则 } T = \frac{\lambda}{u} = 2.5 \text{ s, } \omega = \frac{2\pi}{T} = \frac{4\pi}{5}$$

$$k = \frac{2\pi}{\lambda} = 10\pi$$

(1) $\varphi = \frac{\pi}{2}$

$$y_0 = 0.4 \cos\left(\frac{4\pi}{5}t + \frac{\pi}{2}\right)$$

(2) $y(z, t) = A \cos(\omega t - kz + \varphi_0)$
$$= 0.4 \cos(0.8\pi t - 10\pi z + \frac{\pi}{2})$$

(3) $y_p = 0.4 \cos\left(\frac{4\pi}{5}t + \frac{\pi}{2}\right)$

(4) $y_a = 0.4 \cos\left(\frac{4\pi}{5}t + \frac{\pi}{3}\right)$

$$y_b = 0.4 \cos\left(\frac{4\pi}{5}t - \frac{\pi}{3}\right)$$

15.7



(1) O点: $\varphi_0 = \frac{\pi}{3}$

$$A \text{ 点: } \varphi_A = 0$$

$$B \text{ 点: } \varphi_B = \frac{3\pi}{2}$$

$$C \text{ 点: } \varphi_C = \frac{\pi}{2}$$

(2) $\varphi_0 = -\frac{\pi}{3}$

$$\varphi_A = 0$$

$$\varphi_B = \frac{\pi}{2}$$

$$\varphi_C = \frac{3\pi}{2}$$

15.11

(1) $\bar{I} = \frac{1}{2} \rho A^2 \omega^2 u$

$$\omega = 2\pi v = 2\pi \times 300 = 600\pi$$

$$\therefore \bar{I} = \frac{1}{2} \times 800 \times (1.0 \times 10^{-4})^2 \times (600\pi)^2 \times (1.0 \times 10^{-3})$$

$$= 1440\pi^2 \text{ W/m}^2$$

(2)

$$E = \bar{I} \cdot S \cdot t$$

$$= 1440\pi^2 \times 4 \times 10^{-4} \times 60$$

$$= 341 \text{ J}$$

5.17 $f = 100 \text{ Hz, } \Delta\varphi_0 = \pi, u = 400 \text{ m/s, } AB = 30 \text{ m}$

$$\lambda = \frac{u}{f} = \frac{400}{100} = 4 \text{ m}$$

(1) $\Delta r = |r_2 - r_1| = |30 - 2z| = 2z = 4k$

$$30 - 2z = 4k$$

$$z = 15 - 2k$$

$$z = 1, 3, 5, \dots, 29$$

(2) $\Delta r = AB = 30 \text{ m}$

$$\frac{\Delta r}{\lambda} = \frac{30}{4} = 7.5$$

$$\text{即 } \Delta r = 7.5\lambda = (7 + \frac{1}{2})\lambda$$

说明相位差为 16π 故 AB 连线外任一点的振幅均为 0.30



15.18 (1) S_1 在 100m .

$$r_2 - r_1 = \frac{\lambda}{4} \Rightarrow r_1 - r_2 = -\frac{\lambda}{4}$$

$$\Delta\varphi = \frac{\pi}{2} - \frac{2\pi}{\lambda}(-\frac{\lambda}{4}) = \pi.$$

$$A = 2A_0 \left| \cos\left(\frac{\pi}{2}\right) \right| = 0.$$

(2) S_2 在 100m .

$$\Delta\varphi = \frac{\pi}{2} - \frac{2\pi}{\lambda}(\frac{\lambda}{4}) = 0.$$

$$A = 2A_0 \left| \cos(0) \right| = 2A_0.$$

(3) S_1 与 S_2 在 100m .

$$\Delta\varphi = \frac{\pi}{2} - \frac{2\pi}{\lambda}(2\lambda - \frac{\lambda}{4}) = \pi - \frac{4\pi x}{\lambda}.$$

$$A = 2A_0 \left| \cos\left(\frac{\pi}{2} - \frac{2\pi x}{\lambda}\right) \right|$$

15.20

$$\omega = 600\pi, \quad u = 330\text{m/s}$$

$$\lambda = \frac{u}{\nu} = 1.1\text{m}.$$

$$d = AB = 1\text{m}.$$

设 AC 距离为 x .

$$r_1 = x, \quad r_2 = \sqrt{AB^2 + AC^2} = \sqrt{1+x^2}.$$

$$\therefore \Delta r = r_2 - r_1 = \sqrt{1+x^2} - x$$

$$\Delta r = \frac{\lambda}{2} = 0.55\text{m}$$

$$\therefore \sqrt{1+x^2} - x = 0.55$$

$$\text{解得 } x \approx 0.634\text{m}$$

15.22

$$(1) \nu = \frac{\omega}{2\pi} = 2\text{Hz}.$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{\pi} = 2\text{m}.$$

$$u = \lambda \nu = 4\text{m/s}.$$

$$(2) y = y_1 + y_2$$

$$= 0.06 \cos(\pi z - 4\pi t) + 0.06 \cos(\pi z + 4\pi t)$$

$$= 0.12 \cos(\pi z) \cos(4\pi t)$$

$$A(z) = |0.12 \cos(\pi z)|$$

$$A_{\max} = 0.12\text{m}.$$

$$\pi z = k\pi \Rightarrow z = k$$

$$z = 0, \pm 1, \pm 2, \dots \text{处}.$$

$$A_{\min} = 0\text{m}.$$

$$\pi z = (k + \frac{1}{2})\pi.$$

$$\text{即 } z = k + 0.5.$$

$$z = \pm 0.5, \pm 1.5, \dots \text{处}.$$

15.23

$$y = y_1 + y_2 = 2A \cos(kx) \cos(\omega t)$$

$$(1) A = \frac{0.02}{2} = 0.01\text{m}.$$

$$u = \frac{750}{16} = 46.875\text{m/s}$$

$$(2) \lambda = \frac{2\pi}{k} = \frac{\pi}{8}\text{m}.$$

$$\Delta z = \frac{\lambda}{2} = \frac{\pi}{16}\text{m}$$

$$(3) v = -15 \cos(16z) \sin(\pi \omega t)$$

$$= -15 \cos(16z) \sin(15)$$

$$z = -10.42\text{m/s}$$

$$|v| \approx 10.42\text{m/s}$$

